INCH-POUND

MIL-T-87985
13 March 1989

MILITARY SPECIFICATION
TRENCHER, SELF-PROPELLED, DIESEL-ENGINE-DRIVEN, 70 HP., PNEUMATIC TIRED, WITH OPTIONAL ATTACHMENTS

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 SCOPE. This specification covers a self-propelled diesel-engine-driven pneumatic tired trencher.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-831	Test Reports, Preparation of
MS-51336	Lunette and Coupler, Drawbar, Ring

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3805

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2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

29CFR 1926.600	Equipment
29CFR 1926.602	Materials Handling Equipment
29CFR 1926.1000	Rollover Protective Structures (ROPS) for Material Handling Equipment
29CFR 1926.1002	Protective Frame (ROPS) Test Procedures and Performance Requirements for Wheel-Type Agriculture and Industrial Tractors Used In Construction.

(The Code of Federal Regulation (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents US Government Printing Office Washington DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

THE TIRE AND RIM ASSOCIATION, INC. YEAR BOOK

(Application for copies should be addressed to: The Tire and Rim Association, Inc, 3200 West Market Street, Akron OH 44313.)

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington DC 20591.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Inc, 400 Commonwealth Drive, Warrendale Pennsylvania 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 First Article Test. The supplier shall furnish a complete trenching machine for first article testing in accordance with Section 4 and 6.

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3.2 Design. The trencher shall be pneumatic tire mounted, four wheel drive, four wheel power steer for rigid frames or two wheel steer for center articulated frames, diesel-engine-driven, and be designed to perform underground construction. The machine shall be compatible with optional attachments for trenching, backfilling, plowing, spoils conveyance, rock cutting, and backhoe operations. At least one axle shall be equipped with an automatic locking or limited slip differential.

3.3 Reliability. The trencher, accessories and attachments shall be essentially the current standard products of a well established manufacturer, differing there from only in respects necessary to meet the requirements of this specification.

3.4 Material. The trencher and all parts thereof shall be made of materials of no less quality than those conforming to current engineering and manufacturing practices. Materials shall be free from defects and suitable for the intended service.

3.5 Construction. The trencher shall be complete so that it can be used for any operation specified herein. Construction shall be free from any characteristics or defects that will prevent the machine or associated equipment from passing any of the examination(s) and tests in Section 4.

3.6 Temperature conditions. Without preheating from an external source, the engine shall start within 2 minutes and shall be capable of operating under Full Load conditions within 10 minutes after start, in any ambient temperatures from plus 20°F to plus 115°F without the use of starting aids. When specified (see 6.2), the units shall be equipped with starting aids to permit starting down to minus 20°F. All units shall be capable of operation down to minus 20°F and shall be able to withstand storage temperatures of minus 40°F.

3.7 Use conditions. The trenching machine equipped with trencher and backhoe but without other attachments shall negotiate 30% side slopes and 30% longitudinal slopes or any combination thereof without any wheel of the trenching machine losing contact with the ground. The engine and other pressure lubricated components shall maintain constant operating oil pressure and operate without malfunction under these slope conditions. The trencher and specified attachments shall withstand the usage encountered in normal industrial operations such as trenching, backhoe, plowing and backfilling without malfunction, damage or permanent deformation.

3.8. Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically, shall be insulated, enclosed or guarded. All moving parts which are of such nature or so located as to be a hazard to operating personnel shall be enclosed or guarded. All guards shall be designed to provide access to guarded parts. Protective devices shall not impair the operational functions. The trencher shall comply with 29CFR 1926.600, 29CFR 1926.602, 29CFR 1926.1002. The Rollover Protection Structure (ROPS) shall be of the four (4) post type in accordance with 29 CFR 1926.1000.

3.9 Maintainability. All maintenance described in this specification shall be capable of being performed by one mechanic using common hand tools. Design and construction of the trencher and attachments shall permit ready accessibility to all parts for maintenance and repair in the field. Access shall be in accordance with SAE J925 minimum access dimensions. Replacement and adjustment of component parts and accessories shall be accomplished with minimum drainage of fluids and disturbance to other parts.

3.9.1 Filling, draining, and checking provisions. All fluid systems shall possess a convenient means of filling, draining, and checking the level of the system. All drains shall conduct fluid directly to the ground in a stream with out spilling onto other components. Fluid level indicators or full caps shall identify the type of fluid, state whether the level should be checked cold, hot, or with engine running, and identify appropriate add and full levels.

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3.9.2 Lubrication points. Lubrication points shall be provided for all surfaces requiring lubrication. Pressure relief fittings shall be supplied where the use of pressure lubrication equipment may damage seals. Lubrication fittings shall be in accordance with SAE J534 and be located so as to be clearly visible and easily serviced with a grease gun with a 10-inch flexible hose.

3.9.3 Maintenance operations list. Any four or more operations shall be able to be performed by one person (unless otherwise stated), in the sum of times allotted for each operation.

a. Remove, replace, and adjust all engine driven belts - 1 hour.

b. Remove and replace voltage or current regulator - 15 min.

c. Remove and replace alternator and adjust alternator belts - 30 min.

d. Remove and replace all filter elements, screens, and strainers on the hydraulic system - 30 min.

e. Drain cooling system, remove and replace engine cooling hoses, and refill cooling system - 1 hour.

f. Drain engine lubricating oil, remove and replace oil filter element, and refill crankcase - 30 min.

g. Remove and replace fuel filter elements - 30 min.

h. Remove and replace transmission fluid filter elements and strainer - 45 min.

i. Drain and refill transmission and torque converter - 20 min.

k. Disconnect battery leads, remove and replace batteries, and reconnect battery leads - 15 min.

m. Remove an installed standard chain and replace with an alligator chain - 1 hour.

n. Remove an installed trencher attachment and replace with a rock saw attachment - two mechanics in 3 hours.

3.10 Frame. The frame shall be designed to withstand the maximum stress encountered in operating conditions. It shall be capable of providing adequate support and strength to operate all attachments designed as mounted equipment and approved by the trenching machine's manufacturer.

3.11 Engine. The engine shall be heavy duty air or liquid cooled, either two or four stroke cycle diesel. The engine shall be equipped with an efficient fuel injection mechanism, fuel filter system, air cleaner of the oil bath or dry type filter, and a lubricating oil filter. It shall be capable of producing sufficient horse power to perform all operational requirements specified herein. As a minimum, the engine shall have a Net Brake Horsepower, obtained and corrected in accordance with SAE J1349, of 60.

3.11.1 Engine lubricating system. The engine shall have a lubrication system capable of providing constant lubrication under all normal operating conditions. The engine oil filter shall be a disposable type installed on the pressure side of the engine oil pump.

3.11.2 Engine governor. The engine shall be a variable speed centrifugal type, and shall be either gear driven or shaft driven from the engine. A hand governor control for regulating the governed speed setting throughout the entire engine load range shall be provided and mounted conveniently to the operator.

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3.11.3 Engine cooling system. The cooling system shall have sufficient forced air flow for air cooled engine or shall have sufficient coolant, radiator air flow and pump capacity for liquid cooled engines to properly cool the engine under all operating conditions specified herein.

3.11.4 Exhaust system. The engine shall be equipped with the manufacturer's current standard exhaust system. When an upright exhaust outlet is furnished, it will be of sufficient height that the exhaust gasses will not be routed into the operator cab or compartment. The design of the exhaust system shall prevent the entry of rain.

3.11.5 Engine starting and electrical system. The manufacturer's standard power starting system shall be furnished. Electrical system shall be 12 volts with a fused or circuit breaker system protection and 35 AMP minimum capacity alternator. As a minimum the following lighting shall be furnished:

a. A rotating amber warning beacon or flashing strobe light mounted on top of the rops structure.

b. Instrument lighting.

c. Minimum of two (2) working flood lights which can be adjusted to working areas.

3.11.6 Instrumentation and controls. As a minimum, the following list of instruments shall be furnished and will be located on the dash panel in full view of the operator when in the normal operating position.

a. Ammeter or battery charger indicator.

b. Engine coolant temperature gage/over-heating warning system or overheating protection system.

c. Engine oil-pressure gage or low-oil pressure warning system.

d. Hydraulic-pressure gage (sensing in the crowd circuit).

e. Engine tachometer.

f. Engine hourmeter.

3.12 Service brake. The trencher shall be equipped with either disc-type or shoe-type hydraulic brakes. The brakes shall be capable of stopping the trencher, (with attachments mounted) within a distance of 18 feet from a speed of 8 MPH on dry, level, brushed concrete. Hydrostatic drive systems which use hydraulics for braking are acceptable in lieu of brake shoes or discs.

3.13 Drivetrain. The drivetrain shall provide means of controlling the speed of the vehicle in the forward and reverse direction. A system shall be provided which will protect the vehicle components from damage in the event that the trencher chain hangs while digging. The vehicle's travel shall be independently controlled, thus permitting separate regulation of travel and working speeds as conditions vary.

3.13.1 Speeds. The forward speeds shall permit a travel speed range from 0 miles per hour (MPH) to 8.0 MPH minimum and a reverse transit speed range from 0 MPH to 1.0 MPH minimum. The crowd speed range of the trencher shall be 0 feet per minute (FPM) to 32 FPM minimum.

3.13.2 Clutch. If used, a spring-loaded, foot-operated friction-type clutch and a neutral safety switch shall be incorporated into the system to prevent starting of the engine while in gear.

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3.14 Fuel tank. The fuel tank shall be of sufficient capacity to permit 8 hours operation as specified in 6.1 without refueling.

3.15 Steering. Steering shall be either front and rear axle, or articulated power-assisted, using double-acting hydraulic cylinders and front axle. The system shall be capable of making a 30° turn in either direction and shall crab steer.

3.16 Hydraulic system. The hydraulic system shall include pump(s) hydraulic oil cooler, double action cylinders, strainers, reservoirs, pressure relief valves, control valves, and all necessary lines and fittings as applicable. Means shall be provided in the hydraulic system for bleeding all air trapped in the system. Pressure release check valve(s) will be provided to prevent lowering of lift device when engine is shut down. The system shall provide power for the trenching boom, backfill blade, steering, backhoe operation, and trenching travel speed drive motor. Hydraulic pump capacity available for these functions shall not be less than 15 gallon per minute at a relief valve setting of 1900-2300 PSI. Reservoir capacity shall be 11.5 gallons minimum. Hydraulic circuit shall include a full flow filter system with a replaceable filter cartridge and a means to cool the hydraulic fluid. Hydraulic cylinders shall be of the double action type. All hydraulic controls shall be located within easy reach of the operator and be positioned so as not to create a safety hazard when in use.

3.17 Tires. Tires for the trencher shall be bar lug, high flotation tube or tubeless type, of high commercial quality. Tires shall have individual rated load carrying capacities equal to the maximum individual tire loading in accordance with the TRA Yearbook; however, the minimum size shall not be less than 31 X 15.50-15, 8 ply, flotation type.

3.18 Attachments. All trenchers shall be equipped with attachments a. (or f.), b., and c. When specified (see 6.2), attachments d. through f. shall be furnished. The equipment shall be designed so that the attachments can be changed with common hand tools. Dry-break quick-disconnect hydraulic couplers will be used on hydraulic lines leading to removable attachments.

a. Center-mounted trencher assembly, with crumber attachment

b. Digging chains

c. Backfill blade

d. Backhoe (optional)

e. Rock and concrete saw (optional)

f. Vibratory plow and cable chute assembly (optional)

g. Front-mounted reel carrier assembly (optional)

h. Offset trencher assembly, with crumber attachment (optional, in lieu of item a.).

3.18.1 Center-mounted trencher assembly. The center mounted trencher shall be mechanically or hydrostatically driven with dual, open-end spoils disposer, a mechanical or hydraulic crumber and dual cylinder lift arrangement. The steel digging boom shall be of sufficient length to permit digging to a minimum 48 inch depth. Boom shall be equipped with a digging chain tension adjustment. Boom will be of the wear plate or channel type, equipped with a slip clutch designed to accept standard chains and rock and frost chains.

3.18.2 Digging chains. Standard digging chains shall be a heat-treated, steel roller type with a minimum tensile strength of 70,000 pounds. Standard chain cutting teeth will be bolt-on type. Rock and frost chains shall have a minimum tensile strength of 110,000 pounds. Rock and frost chain cutting teeth

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will be of the pop-out type, tungsten carbide capped. The cutting teeth pattern shall be of the manufacturer's standard design. The cutting patterns on both the standard chain and the rock and frost chain shall be adjustable to dig a trench 8-14 inches wide. Unless otherwise specified (see 6.2), one standard and one rock and frost chain shall be furnished. Each chain shall be furnished with enough bolt-on teeth or teeth holder plates, etc. to make up a chain which will dig any standard width between 8 and 14 inches (8 and 12 inches if unit is to receive offset trenching attachment in accordance with 3.18.8). The factory set up for delivery shall be as specified in 6.2. A tooth pop-out wrench shall be furnished with each unit furnished with a rock and frost chain.

3.18.3 Backfill blade. The trencher shall be equipped with a 6-way power control blade 72 inches (min) in width X 14 inches (min) in height, with a bolt-on, reversible type cutting edge(s). The blade shall also be wider than the maximum trencher width, so as to allow a cut wide enough for the trencher to travel in. The minimum lift height above ground shall be 12.6 inches; minimum blade drop below ground - 3 inches; and minimum blade angles right and left with the direction of travel shall be 38°. Lifting force shall be 1800 pounds (min). Controls shall be conveniently located for operation from the operator's seat.

3.18.4 Backhoe. When specified (see 6.2), the trencher shall be equipped with a backhoe. The backhoe shall be hydraulically operated, complete with seat, controls, connecting lines and fittings, outriggers, and bucket. The backhoe module shall be compatible with the trencher and shall attach to the vehicle so as not to interfere with backfilling or trenching operations. The operator's seat shall be positioned to provide a clear view of the work area. Controls shall be conveniently accessible to the operator when seated. Outriggers shall be hydraulically operated and have reversible street pads. The backhoe shall be equipped with a 12 inch (min) bucket with standard teeth. Swing arc of the backhoe shall be 180° (min). Bucket rotation shall be 175° (min). Digging depth shall be 8 feet (min). Lift capacity shall be 1000 lbs (min). Reach from center of rotation shall be 136 (min). Material loading height shall be 98 inches (min). Digging force of bucket cylinder shall be 4,000 lbs (min).

3.18.5 Rock and concrete saw. When specified (see 6.2), the trencher shall be equipped with a rock saw. The rock saw shall be compatible with the trencher and easily attached or removed. It shall be of the disc/wheel type, heavily reinforced throughout to reduce stress; hydraulically or mechanically driven; capable of trenching effectively in coral, rock, frost and paved surfaces to a width of 4 inches (min) and depth of 30 inches (min). Standard attachments shall consist of adjustable stabilizers to aid in positive depth control and reduce bounce. Water spray attachments to be provided and positioned to direct water for cooling the cutting area. Bolt-on adapter/plates with pop-out, free rotating, carbide tip teeth with effective cutting arrangement shall be furnished. The saw module shall be equipped with stabilizer feet/stands for positioning the saw upright when disconnected and stowed.

3.18.6 Vibratory plow and cable chute assembly. When specified (see 6.2), a vibratory plow shall be furnished capable of plowing a trough (trench) 30 inches deep and wide enough to bury a 2 inch cable. The plow shall be of the shaker type, hydrostatically or mechanically driven. A removable cable chute shall be provided to permit burial of cable up to 2 inches in diameter. Additional chutes shall be supplied as specified in 6.2.

3.18.7 Front-mounted cable carrier. When specified (see 6.2), a front mounted cable reel carrier capable of handling reels up to 32 inches wide X 48 inches in diameter will be furnished. The carrier shall be compatible with the trencher and have a lift capacity of 1500 pounds minimum. A check valve to prevent lowering carrier when engine is not running shall be furnished. A removable cable guide for routing cable over the power unit to the plow shall be furnished.

3.18.8 Offset trencher assembly, with crumber attachment. When specified (see 6.2) an offset trencher assembly shall be provided in lieu of the standard center-mounted trencher assembly. The performance specifications shall be the same as specified in 3.18.1 except trench width shall be 8 - 12 inches and only a single spoils auger is required.

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3.19 Trailer. When specified (see 6.2), a trailer shall be provided for the trencher. The trailer shall be tandem axle, flatbed unit with beaver tail and loading ramps. The trailer shall be fitted with a lunette in accordance with MS 51336. The lunette height shall be adjustable from 22 to 40 inches in three inch or smaller increments. Height adjustment shall be possible by two mechanics using common hand tools in 15 minutes or less. The trailer shall be equipped with either a full width ramp or adjustable ramps capable of loading wheeled equipment with tires spaced as narrow as 30 inches to as wide as the width of the trailer. The trailer shall have a front mounted jack rated at not less than 30,000 pounds. The trailer shall have an overall length of between 26 and 30 feet and shall be rated for 20,000 pound weight. Design shall permit the trencher to load and offload with a minimum of two inches clearance maintained between the trencher equipped with any attachment (trailer bed level, on hard, level surface). Two inches of clearance above the ground shall also be maintained during on/off loading. There shall be provisions for quickly and effectively securing the trencher to the trailer for transport. Trailer tires shall be in accordance with the Tire and Rim Association Handbook recommendations. The trailer shall be equipped with air brakes. Brakes, lighting, wiring, and rear end protection shall be in accordance with the Federal Motor Carrier Safety Regulations. A seven contact receptacle conforming to SAE 560, with connectors connected and color coded as specified therein, and equipped with a spring loaded cover, shall be provided. The type and color of paint on the trailer shall be the same as that required for the trencher.

3.20 Lubrication. Lubrication means shall be provided for all moving parts of the trencher, attachments, and trailer requiring lubrication. Lubrication fittings shall be in accordance with SAE J534. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts. The trenchers shall be conspicuously tagged to identify the lubricants types and their temperature range.

3.21 Data plates. The following plates shall be installed in a conspicuous location on the end item.

3.21.1 Identification plate. A corrosion resistant metal name plate shall be permanently and legibly marked with the following information, in accordance with MIL-STD-130.

nomenclature
manufacturer's model number
manufacturer's serial number
special characteristics (HP, RPM, etc.)
national stock number
manufacturer's name
date of manufacture
warranty expiration date
curb weight and CG location

3.21.2 Lubrication plates. A corrosion resistant metal lubrication plate(s) shall illustrate the lubrication points on the vehicle and shall identify the type of lubricant and service interval for each point.

3.21.3 Warning plates. Warning plates shall identify potentially hazardous conditions, locations and actions. Plates shall be sized and located so as to be conspicuous to the person at risk.

3.21.4 Instruction plates. All controls shall be labeled to identify the location and function of each control position. Instructions cast or stamped on or beside the controls are acceptable.

3.22 Treatment and painting. Mix trucks shall be painted in accordance with standard commercial practice provided that the paint and primer is both lead and chromate free. Color shall be 24052 of FED-STD-595 unless otherwise specified in 6.2. The exterior finish shall show no evidence of uneven application, runs, sags, curtains, orange peel, lack of adhesion, or other defect. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted to match the chassis.

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3.22.1 Markings. Markings (US Air Force and Registration Numbers) will be applied on dark green vehicles utilizing black paint color number 37038. Place registration numbers on each side of vehicle and on rear if space permits. Size of letters and numerals to be used are as follows.

US AIR FORCE 1-1/2 INCHES
REGISTRATION NUMBER 1-1/2 INCHES
REG NUMBER SPACING EXAMPLE 77D1977

Lifting and tie down locations shall be identified by 1-1/2 inch letters stating "LIFT HERE" and "TIE DOWN HERE" as applicable.

3.23 Workmanship.

3.23.1 Steel fabrication. Steel used in fabrication of equipment shall be free from kinks and sharp bends. Straightening of material shall be accomplished by methods that will not cause injury to the metal. Shearing and clipping shall be done neatly and accurately. All bends of a major character shall be made under controlled means in order to insure uniformity of size and shape.

3.23.2 Bolted connections. Bolt holes shall be accurately punched or drilled, and shall have burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts and screws shall be tight.

3.23.3 Riveted connections. Rivet holes shall be accurately punched or drilled, and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and uniform size for rivets having the same diameters.

3.23.4 Welding. Welding shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter, weld penetration shall be such as to provide transference of maximum stress through the base metal juncture.

3.23.5 Castings. Castings shall be sound and free from patching, misplaced coring, warping or other defects which might render the casting unsound for use.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other Government approved facilities suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements of the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Notification of testing. The Government shall be, in accordance with contract provisions, notified of all testing to be performed in fulfillment of the requirements of this specification so that arrangements

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may be made to witness the testing. Addresses and specific requirements of notifications shall be provided in the solicitation, contract, or by the Government's contracting officer upon request.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article test

b. Quality conformance inspection.

4.2.1 First article test. First article inspection shall be performed on the first article sample submitted in accordance with 3.1. Failure of the first article sample to pass the examination of one or more tests specified shall be cause for rejection. First article inspection shall consist of the examination in 4.4.1, and 4.6, and tests in 4.4. Tests on the optional attachments (4.4.8, 4.4.10 - 4.4.13) need only be performed if the attachments are being provided under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on each machine prior to being offered for acceptance under the contract. Failure of the machine to pass the examination of one or more tests shall be cause for rejection. Quality conformance inspection shall consist of the following:

a. Examination (4.4.1)

b. Test (4.5)

c. Inspection of preservation, packaging, packing and marking for shipment and storage (4.6).

4.3 Test conditions.

4.3.1 Apparatus. The apparatus used in conjunction with the testing specified herein shall be of laboratory precision and shall be calibrated at proper intervals to ensure accuracy.

4.3.2 Preparation for tests. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, any of which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any fluid, including fuel, coolant, lubrication or hydraulic fluid, under any operation condition except as allowed herein.

c. Structural failure, evidence of impending failure, or permanent deformation of any component or joint.

d. Evidence of excessive wear.

e. Interference between vehicle components or between the vehicle and the ground (including all required obstacles).

f. Misalignment of any component.

g. Evidence of undesirable roadability characteristics, including instability in handling while cornering, braking, and traversing rough terrain as required.

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h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

4.4 Tests.

4.4.1 Examination. Each trenching machine, including the preproduction sample, shall be examined for compliance with the requirements specified in Section 3 of the specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failures to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examination and dimensional measurements. Non-compliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4.2 Temperature conditions test. The ability of the trencher to meet the requirements of 3.6 at both extreme temperatures of plus 20°F and plus 115°F shall be demonstrated. If optional starting aids are specified, the ability to start at minus 20°F shall be demonstrated. In lieu of this test, the supplier, at this option, may furnish a notarized certification that his trencher will either meet or exceed the requirements of 3.6. Failure to comply with the specified requirements or to furnish the optional notarized certificate shall be cause for rejection.

4.4.3 Use conditions test. The trenching machine shall be run on 30 percent side slopes, 30 percent longitudinal slopes, and combinations thereof. Loss of wheel contact with ground, a reduction or inconsistent oil pressure, or failure of engine, oil system and attachments to operate as on flat terrain shall be cause for rejection.

4.4.4 Maintenance testing. After having performed the inspection according to 4.4.1, an authorized government representative shall select between four and all of the tasks listed in 3.9.3. These tasks shall be performed sequentially by one person (unless otherwise indicated in 3.9.3) in the sum of the time allotted for each task. All tools, parts, and supplies shall be on hand at the start of the test. The list of tasks performed and the total time taken to perform the tasks shall be recorded. The use of any tools other than common hand tools shall be noted. Power tools shall not be used.

4.4.5 Speeds. The capability of the trencher to meet the forward, reverse, and crowd speed in 3.13.1 and the independent regulation of travel and working speeds shall be demonstrated.

4.4.6 Service brakes. The trencher, with backfill blade, backhoe and center digging assembly mounted, shall be run at speed of 8 mph on level, brushed concrete surface and the service brakes applied. Service brakes stop the trencher within a distance of 18 feet from the point of application.

4.4.7 Steering. Ability of the trencher to make minimum 30° turns to the direction of travel shall be demonstrated. Failure shall be cause for rejection.

4.4.8 Center mounted and offset trencher test. Each trenching assembly shall dig a trench 8 inches wide, 48 inches deep, and 50 feet long in loam clay soil with the standard chain at maximum speed. Disposition of spoils and operation of the crumber shall be observed. Failure of the trenching attachment to meet requirements in para 3.18.1 or 3.18.8 will be cause for rejection. Test will demonstrate proper slip-clutch operation and accessibility for adjustment and repair if mechanical drive is used.

4.4.9 Back fill test. The trencher shall backfill a trench 50 feet long. All functions specified in 3.18.3 shall be verified in an operational test.

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4.4.10 Backhoe test. The backhoe shall dig a square hole 4 feet by 4 feet by 8 feet deep. Backhoe shall lift 1000 pounds suspended from the outside of the bucket to a height of 4 feet.

4.4.11 Rock and concrete saw test. The saw shall be tested for compliance with 3.18.5 and to cut a 4 inch wide trench X 30 inches deep by 10 feet long in a concrete surface within three hours with a minimum of bounce. The adjustable stabilizer will demonstrate their effectiveness for positive cutting depth control. Attaching and detaching of the saw will be demonstrated for ease and time required. Means of stowing the saw in the safe and upright position, when detached from trencher, will be demonstrated.

4.4.12 Vibratory plow and reel carrier test. Remove the trenching boom and replace with the plow attachment. The plow shall trench 30 inches deep for a distance of 50 feet. The cable reel will be tested to assure it meets lift capacity as required in para 3.18.7. When used in conjunction with the plow, the lift and cable guard shall permit the movement of 2 inch diameter cable to the plow without scoring or damage to the cable. Failure to perform these requirements in a timely manner shall be cause for rejection.

4.4.13 Trailer test. The trailer shall be attached to a tow vehicle and the trencher, complete with mounted attachments (rock cutter, backfill blade, and backhoe) shall be driven aboard the trailer and tied down for transport. Should sufficient room and weight capacity be available, any optional items such as vibratory plow and reel carrier shall also be placed on the trailer and tied down for transport. The trailer shall be towed over paved roads for a distance of 5 miles at speeds as high as 45 mph, then the trailer shall be towed a distance of 3 miles over unimproved roads at speeds up to 10 mph. The trailer shall track the towing vehicle without weaving from side to side or exhibiting excessive lateral sway or damage to the trailer, trencher or equipment. The trailer shall be stopped at least six times to demonstrate action of the brakes. All lights (directional, signal, hazard, license plate,, and stop lights) shall be operated. As final test, the rear stabilizer shall be placed in the standing position, and the tongue jack operated to release trailer from tow vehicle. Ramps shall be positioned and trencher shall be driven from trailer to ground. Any damage to the trailer, trencher or equipment or any malfunction shall be cause for rejection.

4.4.14 Test report. Upon completion of the first article testing, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized government representative shall countersign all data sheets and the report. The body of the report shall be organized so as to parallel section four of this specification to the extent possible. Each item in the report shall identify the paragraph of this specification to which it pertains. The following shall be incorporated as a part of the test report:

a. All failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles or position of the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made, or proposed to be made, and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements, that if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. The engine manufacturer's certified power curves, obtained and corrected in accordance with SAE J1349, shall be furnished. The Net Brake Horsepower shall be that determined from the engine manufacturer's certified power curves. Power curves shall show compliance with the requirements of 3.11. Maximum governed engine speed shall be marked on the chart.

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g. The contractor shall furnish the following commercial spec sheets as part of the first article test report.

(1) Engine spec sheet.

(2) Transmission spec sheet.

(3) Chassis spec sheet.

(4) Hydraulic pump and motor spec sheets.

4.5 Production unit operational test. Each trencher produced in fulfillment of the government contract or order shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. All lights shall be turned on for the duration of the test. The engine shall be started and subjected to a warm-up period as recommended by the manufacturer. The trencher shall be given a run-in test, and all controls shall be operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operate properly, fully, and without restriction or malfunction. Unit shall be inspected for the rejection criteria listed in 4.3. Trailers, when provided, shall each be towed at least 0.5 miles. The trailer shall be inspected for brake drag by checking for overheating. After checking for overheating, the brakes shall be engaged twice for 25 mph - 0 mph stops. Brakes shall be checked for proper function. Gladhands shall be removed and the trailer shall be dragged a short distance. All wheels shall remain locked. The brake lines shall be inspected to ensure that they will withstand full suspension travel without being damaged.

4.6 Inspection of preservation, packaging, packing and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage of the trencher, attachments and accessories shall be inspected to verify conformance with the terms of the contract or order.

5. PACKAGING.

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with the terms of the contract.

6. NOTES.

6.1 Intended use. The trencher, attachments and accessories are intended for use in underground construction, such as trenching, plowing, backfilling and backhoe work as required by Air Force Electronic Installation Groups.

6.2 Ordering data. Procurements documents should specify the following:

a. Title, number, and date of this specification.

b. When low temperature starting aids are required (see 3.6).

c. Attachments required (see 3.18).

(1) Backhoe (see 3.18.4).

(2) Rock and concrete saw (see 3.18.5).

(3) Cable plow. Additional cable chutes (specify type and diameter of cable) (see 3.18.6).

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(4) Front-mounted Reel Carrier Assy (See 3.18.7).

(5) Offset trencher assy, with crumber attachment (see 3.18.8).

d. Digging chain requirements, (see 3.18.2).

#	chain type	chain strength (lbs)	factory set-up width	additional segments for width range of (8" min, 14" max)
1	standard *	70,000	8 in.	8 in. - 14 in. for std boom 8 in. - 12 in. offset boom
2	rock/frost	110,000	8 in.	8 in. - 14 in. for std boom 8 in. - 12 in offset boom
3				
4				

* To be installed on the trencher

e. When trailer is required (see 3.19).

f. Paint color, if different than specified (see 3.22).

6.3 First Article. The sample may be a production item or a preproduction item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed item and shall be made or shall have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. Approval of a first article unit shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification. The first article unit may be shipped as part of the contract.

6.4 Subject term (Key Word) listing.

Diesel-engine-driven
Pneumatic tired
Self-propelled
Trencher

Custodians:
Air Force - 99

Preparing activity:
Air Force - 84

Project No: 3805-F132

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